

# BRENDAN HARRIS

Postdoctoral Research Associate



brendanjohnharris.github.io



0000-0003-3412-4186



brendan.harris@sydney.edu.au



Sydney, Australia



brendanjohnharris



Brendan Harris



brendanjohnharris



+61 466 956 165

## SUMMARY

Postdoctoral Research Associate in the Complex Systems Group at the School of Physics, The University of Sydney. Interested in understanding the dynamical mechanisms of computation and communication in the brain, especially for visual processing. Use new analytic tools, dynamical systems theory, large-scale data, and biophysical neural-circuit models for a multi-level understanding of brain function.

## RESEARCH INTERESTS

### *Circuit- and systems-level neuroscience*

Spiking neural circuits; mean-field models; hierarchical processing; oscillations and traveling waves; visual system.

### *Bridging statistical physics and neural dynamics*

Time-series analysis; stochastic processes; criticality; cross-scale dynamics; dimensionality reduction; open software.

## CONTENTS

### Education

### Publications and Preprints

### Conferences

### Workshops and talks

### Research experience

### Supervision

### Teaching

### Industry

### Awards

### Referees

## EDUCATION

|             |                                                                                                                                     |                          |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 2022        | <b>Physics PhD Candidate (Submitted)</b><br><i>Cross-scale dynamics in the working regime of the visual cortex</i>                  | The University of Sydney |
| 2021        | <b>Physics Honours, Class I and the University Medal</b><br><i>Inferring parametric variation across non-stationary time series</i> | The University of Sydney |
| 2018 – 2020 | <b>Bachelor of Science/Bachelor of Advanced Studies (Dalyell Scholar)</b><br>Majored in Physics and Neuroscience                    | The University of Sydney |

## PUBLICATIONS AND PREPRINTS

|          |                                                                                                                                                                                                                                                                                                                                                                                                                |                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 2026     | <i>Anomalous dynamics in the working regime of the visual cortex</i><br>Brendan Harris and Pulin Gong                                                                                                                                                                                                                                                                                                          | In preparation                                           |
| 2026     | <i>Nested spatiotemporal theta–gamma waves organize hierarchical processing across the mouse visual cortex</i>  <br>Brendan Harris and Pulin Gong                                                                                        | <a href="#">Nature Communications</a><br>vol. 17, pp.    |
| Jul 2025 | <i>Canonical time-series features for characterizing biologically informative dynamical patterns in fMRI</i>  <br>Imran Alam, <b>Brendan Harris</b> , Patrick Cahill, Oliver Cliff, Marija Markicevic, Valerio Zerbi, and Ben D. Fulcher | <a href="#">Aperture Neuro</a><br>vol. 5, pp.            |
| Aug 2024 | <i>Tracking the Distance to Criticality in Systems with Unknown Noise</i>  <br>Brendan Harris, Leonardo L. Gollo, and Ben D. Fulcher                                                                                                 | <a href="#">Physical Review X</a><br>vol. 14, pp. 031021 |

|          |                                                                                                                                                                                                                                                                                                       |                                                                  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| May 2024 | <b><i>Distributed and dynamical communication: a mechanism for flexible cortico-cortical interactions and its functional roles in visual attention</i></b> <br>Shencong Ni, <b>Brendan Harris</b> , and Pulin Gong | <b>Communications Biology</b><br>vol. 7, pp. 550                 |
| Jul 2021 | <b><i>Approximate Modal Cut-Off Wavelengths and the V-Parameter for M-type Optical Fibers and Its Novel Applications</i></b><br>Deepak Jain, Mark A George, <b>Brendan Harris</b> , and Simon Fleming                                                                                                 | <b>Journal of Lightwave Technology</b><br>vol. 39, pp. 4478–4488 |

## CONFERENCES

|          |                                                                                                                                                                                                                                                                                   |                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Jun 2025 | <b>OHBM 2025</b> <br>Brisbane, QLD, Australia<br><i>2072: Nested spatiotemporal dynamics organize hierarchical processing in the mouse visual cortex</i><br><b>Brendan Harris</b> and Pulin Gong | Poster            |
| Jun 2025 | <b>EPC/APCV 2025</b><br>The University of New South Wales<br><i>Spatiotemporal theta–gamma waves organize hierarchical processing in the mouse visual cortex</i><br><b>Brendan Harris</b> and Pulin Gong                                                                          | Talk              |
| Mar 2025 | <b>COSYNE 2025</b> <br>Montreal, QC, Canada<br><i>1–117: Tracking the distance to criticality across the mouse visual hierarchy</i><br><b>Brendan Harris</b> , Leonardo Gollo, and Ben Fulcher   | Poster            |
| Feb 2025 | <b>NeuroEng 2025</b><br>The University of Melbourne<br><i>Nested spatiotemporal theta–gamma waves organize hierarchical visual processing</i><br><b>Brendan Harris</b> and Pulin Gong                                                                                             | Talk              |
| Jul 2024 | <b>2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)</b><br><i>1–4: Default Mode Network Detection using EEG in Real-time</i><br>Navin Cooray, Chetan Gohil, <b>Brendan Harris</b> , Shaun Frost, and Cameron Higgins      | Conference paper  |
| Oct 2023 | <b>IBRO Neuroscience Reports (IBRO World Congress 2023)</b><br>Granada, Spain<br><i>S783: Nested and non-stationary oscillatory bursts underlie flexible inter-areal neural communication</i><br><b>Brendan Harris</b> and Pulin Gong                                             | Poster            |
| Jul 2022 | <b>31st Annual Computational Neuroscience Meeting: CNS*2022</b><br>Melbourne, Australia<br><i>P36: Summarizing non-stationarity in spatio-temporal neural data</i><br><b>Brendan Harris</b> and Ben Fulcher                                                                       | Poster            |
| Jun 2022 | <b>28th Annual Meeting of the Organization for Human Brain Mapping</b><br><i>Summarizing non-stationarity in spatio-temporal neural data</i><br><b>Brendan Harris</b> and Ben Fulcher                                                                                             | Virtual Poster    |
| Sep 2020 | <b>Frontiers in Optics / Laser Science (2020)</b><br><i>FM4D.2: Approximate normalized frequency (V-parameter) and modal cut-off wavelengths of M-type optical fibers</i><br>Deepak Jain, Mark George, <b>Brendan Harris</b> , and Simon Fleming                                  | Oral presentation |

## WORKSHOPS AND TALKS

---

|          |                                                                                                                                                                                                                                                   |                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Apr 2026 | <b>Maths in the Brain</b> <br><i>Anomalous dynamics in the working regime of the visual cortex</i><br>Prize: "Best ECR talk"                                     | Talk              |
| May 2025 | <b>Center for Neuroscience Imaging Research (CNIR), IBS, Republic of Korea</b><br><i>Nested spatiotemporal <math>\theta</math>-<math>\gamma</math> waves organize hierarchical visual processing</i><br>Invited to give a talk on my PhD research | Talk              |
| Mar 2025 | <b>Pre-Cosyne BrainHack 2025</b><br>Montreal, QC, Canada<br>Workshop on open science and open-source tools for computational and systems neuroscience.                                                                                            | Workshop attendee |
| Aug 2024 | <b>Strong Compute Chess AI Hackathon</b><br>Sydney, NSW, Australia<br>Hackathon on transformer models and distributed GPU training for chess                                                                                                      | Workshop attendee |
| Aug 2024 | <b>Emerging Aspirations in Complex Systems</b><br>The University of Sydney<br><i>Spatiotemporal <math>\theta</math>-<math>\gamma</math> waves in the mouse visual cortex</i><br>Centre for Complex Systems, The University of Sydney              | Workshop talk     |
| May 2024 | <b>School of Physics HDR symposium</b><br>The University of Sydney<br><i>Spatiotemporal <math>\theta</math>-<math>\gamma</math> waves in the mouse visual cortex</i><br>School of Physics, The University of Sydney                               | Workshop talk     |
| Oct 2023 | <b>MIP:Lab, Geneva Biotech Campus</b><br>Geneva, Switzerland<br><i>Burst-based inter-areal neural communication</i><br>Presented my PhD work to the MIP:Lab, hosted by Dr. Enrico Amico.                                                          | Talk              |
| Jul 2023 | <b>Cognitive Neuroscience Hub, The University of Melbourne</b><br>Melbourne University<br><i>Burst-based inter-areal neural communication</i><br>Presented my PhD work, hosted by Dr. Jacob Paul.                                                 | Talk              |
| Jul 2023 | <b>tLab</b><br>Monash University<br><i>Burst-based inter-areal neural communication</i><br>Presented my PhD work to the lab of Professor Nao Tsuchiya.                                                                                            | Talk              |
| Jan 2023 | <b>Complexity, Criticality and Computation Symposium: C3-2023</b><br>Heron Island<br><i>Burst-based inter-areal neural communication</i><br>Symposium held by the Centre for Complex Systems (The University of Sydney).                          | Workshop talk     |
| Nov 2022 | <b>BrainHack Australia 2022</b><br>The University of Sydney<br><i>Brainharmonic</i><br>Contributed to a project aiming to synthesize music from neural signals using deep neural networks.                                                        | Workshop attendee |
| Oct 2022 | <b>OpenScope NeuroDataReHack Hackathon</b><br>Seattle, WA, USA<br>Workshop on re-analyzing open-source neural data, hosted by the Allen Institute                                                                                                 | Workshop attendee |

|          |                                                                                                                                                                                                                                            |               |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Oct 2022 | <b>Yale Medical School</b><br><i>Feature-based analysis of neural time series</i><br>Invited by Dr. Evelyn Lake to present the results of my projects involving time-series features to a journal club.                                    | Talk          |
| Aug 2022 | <b>Emerging Aspirations in Complex Systems</b><br>The University of Sydney<br><i>Inferring parametric variation across non-stationary neural time series</i>                                                                               | Workshop talk |
| Jul 2022 | <b>31st Annual Computational Neuroscience Meeting: CNS*2022</b><br>Melbourne, Australia<br><i>Inferring parametric variation across non-stationary neural time series</i><br>In workshop 'Highly comparative analysis of neural dynamics'. | Workshop talk |
| Dec 2021 | <b>Neuromatch 4.0</b> <br><i>Summarizing non-stationarity in spatio-temporal neural data</i>                                                              | Flash talk    |

## RESEARCH EXPERIENCE

---

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                       |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 2022           | <b>Cross-scale dynamics in the working regime of the visual cortex</b><br><i>Dr. Pulin Gong, Complex Systems Group, The University of Sydney</i><br>In my PhD work, I aimed to understand the principles of computation in spatiotemporal neural systems and their role in transforming complex stimuli into flexible behavior. Specifically, I focus on analyzing spatiotemporal patterns, criticality, cross-scale dynamics, and nonlocal diffusion in mouse electrophysiology data, spiking neural circuits and their mean-field reductions, as well gaze trajectories. | PhD research          |
| Mar – Nov 2021 | <b>Inferring parametric variation across non-stationary time series</b><br><i>Dr. Ben Fulcher, Complex Systems Group, The University of Sydney</i><br>Developed a method to summarize non-stationary dynamics using time-series features and dimensionality reduction, along with a new approach to dimensionality reduction that uses baseline datasets to account for redundant measurements. I applied this method to both simulated data and mouse electrophysiology recordings.                                                                                       | Honours research      |
| Aug – Nov 2020 | <b>Relating BOLD dynamics, interneuron densities, and DREADD activation</b><br><i>Dr. Ben Fulcher, Complex Systems Group, The University of Sydney</i><br>Explored links between BOLD dynamics, neuronal excitability, and cell densities by combining fMRI, DREADD activation, and neuron-count data with time-series features ( <i>hctsa</i> ) and machine-learning classifiers.                                                                                                                                                                                         | Dalyell project       |
| Jan – Mar 2020 | <b>Quantum Control Laboratory</b><br><i>Dr. Cornelius Hempel, School of Physics, The University of Sydney</i><br>Completed lab projects in optics, instrumentation, and Python/Julia/C++ programming, including building an interferometer for ion trap vibration measurements, testing signal generator stability, and developing Arduino-based sensors.                                                                                                                                                                                                                  | Denison project       |
| Aug – Nov 2019 | <b>Classifying the stability of near-critical systems</b> <br><i>Dr. Ben Fulcher, Complex Systems Group, The University of Sydney</i><br>Used machine-learning algorithms and time-series features to classify time series from monostable and bistable regions of dynamical systems.                                                                                                                                                                                                   | Undergraduate project |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                       |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Jul – Aug 2019      | <b>Interneuron densities and intrinsic timescales in the mouse brain</b><br><i>Dr. Ben Fulcher, Complex Systems Group, The University of Sydney</i><br>Investigated the relationship between interneuron subtypes and brain dynamics by analysing fMRI time-series features ( <i>hctsa</i> ) and interneuron density datasets in Matlab and Python.                                                                                                                                        | Casual lab assistant  |
| Mar – Jun 2019      | <b>Numerical analysis of M-type optical fibers</b><br><i>Dr. Deepak Jain, School of Physics, The University of Sydney</i><br>Modelled M-type optical fibres in COMSOL and formulated expressions describing their unique properties, cut-off wavelengths, anticrossings, and mode profiles.                                                                                                                                                                                                | Undergraduate project |
| Aug 2018 – Aug 2021 | <b>Tracking the distance to criticality in systems with unknown noise</b> <br><i>Dr. Ben Fulcher, Complex Systems Group, The University of Sydney</i><br>Searched for time-series features that best predict the distance to criticality in noisy dynamical systems; this involved numerical simulation and time-series analysis ( <i>hctsa</i> ) in Matlab and with the School of Physics HPC cluster. | Undergraduate project |

## SUPERVISION

---

|                |                                                                                   |           |
|----------------|-----------------------------------------------------------------------------------|-----------|
| Mar – Nov 2026 | <b>Jude Metcalf</b><br><i>The University of Sydney</i><br>Physics Honours student | Secondary |
| Mar – Nov 2026 | <b>Murray Jones</b><br><i>The University of Sydney</i><br>Physics Honours student | Secondary |
| Mar – Nov 2025 | <b>Aiden Sloots</b><br><i>The University of Sydney</i><br>Physics Honours student | Secondary |

## TEACHING

---

|                     |                                                                                                                                                                                                                                                                                                                                                     |       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Jul – Nov 2024      | <b>Computational optics</b><br><i>The University of Sydney</i><br>Tutored a computational optics lab                                                                                                                                                                                                                                                | Tutor |
| Feb – Mar 2023      | <b>3rd-year Statistical Mechanics</b><br><i>The University of Sydney</i><br>Tutored a combined lecture and tutorial class on statistical mechanics, covering topics such as the Einstein solids, the Ising model, Peierls argument, Monte-Carlo sampling, phase transitions, criticality, and the renormalization group.                            | Tutor |
| Aug 2022 – Mar 2023 | <b>3rd-year Interdisciplinary Physics computational lab</b><br><i>The University of Sydney</i><br>Tutored the Matlab computational labs of PHYS3888, a course on the physics of complex systems that introduces third year physics students to networks, linear and nonlinear dynamical systems, criticality, and machine learning/neural networks. | Tutor |

|                |                                                                             |               |
|----------------|-----------------------------------------------------------------------------|---------------|
| Mar – Nov 2022 | <b>Second-year physics Matlab course</b><br><i>The University of Sydney</i> | <b>Tutor</b>  |
| Aug – Nov 2018 | <b>Science Dalryell Showcase</b><br><i>The University of Sydney</i>         | <b>Mentor</b> |

## INDUSTRY

---

|                |                                                                                                                                                                                                                                                                       |                           |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Mar – Dec 2023 | <b>Resonait Medical Technologies</b><br><i>Dr. Cameron Higgins</i><br>Worked on a casual basis writing Matlab code to interface with EEG headsets, infer the activation of brain networks, and implement neurofeedback for the monitoring and treatment of depression | <b>Software developer</b> |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|

## AWARDS

---

|      |                                                                                                                                                                        |                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2025 | <b>Postgraduate Research Support Scheme</b><br><i>The University of Sydney</i><br>To fund travel and accommodation for the COSYNE 2025 conference (Montreal, Canada)   | <b>Scholarship</b> |
| 2025 | <b>Cosyne Presenters Travel Grant</b><br><i>COSYNE</i><br>To fund travel and accommodation for the COSYNE 2025 conference (Montreal, Canada)                           | <b>Scholarship</b> |
| 2023 | <b>Postgraduate Research Support Scheme</b><br><i>The University of Sydney</i><br>To fund travel and accommodation for the IBRO2023 conference (Granada, Spain)        | <b>Scholarship</b> |
| 2022 | <b>Dean's List of Excellence in Academic Performance</b><br><i>The University of Sydney</i><br>For academic performance during my Honours year                         | <b>Recognition</b> |
| 2022 | <b>Honours Class I and the University Medal</b><br><i>The University of Sydney</i><br>For academic and research performance during my Honours year                     | <b>Recognition</b> |
| 2022 | <b>Research Training Program (fee offset and stipend)</b><br><i>The University of Sydney</i><br>To fund my PhD studies                                                 | <b>Scholarship</b> |
| 2022 | <b>Emerging Aspirations in Complex Systems Award</b><br><i>The University of Sydney</i><br>To fund accommodation for the C3 symposium                                  | <b>Scholarship</b> |
| 2021 | <b>The University of Sydney Physics Foundation Scholarship No III</b><br><i>The University of Sydney</i><br>For academic performance in 3rd year undergraduate Physics | <b>Scholarship</b> |

|      |                                                                                                                                                                                           |                    |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2021 | <b>University of Sydney Academic Merit Prize</b><br><i>The University of Sydney</i><br>For academic performance during my Honours year                                                    | <b>Scholarship</b> |
| 2019 | <b>Denison Research Scholarship and Denison Relocation Scholarship</b><br><i>The University of Sydney</i><br>To fund a 6-week summer research project with the Quantum Control Laboratory | <b>Scholarship</b> |
| 2018 | <b>Dean's List of Excellence in Academic Performance</b><br><i>The University of Sydney</i><br>For academic performance in my 1st year of undergraduate study                             | <b>Recognition</b> |
| 2017 | <b>Sydney Scholars Award</b><br><i>The University of Sydney</i><br>For academic results in the higher school certificate                                                                  | <b>Scholarship</b> |

## REFEREES

---

### Dr. Benjamin D. Fulcher

School of Physics  
The University of Sydney

Physics Building, Physics Rd  
Camperdown 2006

Ph: +61 481 563 731  
ben.fulcher@sydney.edu.au

### Dr. Pulin Gong

School of Physics  
The University of Sydney

Physics Building, Physics Rd  
Camperdown 2006

Ph: +61 2 9036 9368  
pulin.gong@sydney.edu.au

### Dr. Cameron Higgins

Resonait Medical Technologies

Terrigal, NSW

Ph: +61 423 344 596  
cam@resonait.com